Capital Asset Pricing Model.

- The capital asset pricing model (CAPM) is used to determine a theoretically appropriate required rate of return of an asset, if that asset is to be added to an already well-diversified portfolio (market portfolio).
- The model takes into account the asset's sensitivity to market risk (also known as systematic risk or non-diversifiable risk), often represented by the quantity beta (β) in the financial industry, as well as the expected return of the market and the expected return of a theoretical risk-free asset.

The Market Portfolio

- In our last lecture, we constructed an efficient frontier, capital allocation lines, and the tangency portfolio using two stocks and the risk-free asset
- Suppose that we repeated the exercise with <u>all</u> securities, not just two. We would obtain a tangency portfolio, a capital allocation line (known as the security market line (SML))
- The market portfolio is the value-weighted portfolio of <u>all</u> securities (yes, <u>all</u> of them)...
- ...but in practice, we approximate the returns of the market portfolio with the returns of the S&P 500
- Investors <u>demand</u> a combination of the risk-free asset and the tangency portfolio; the market <u>supplies</u> the market portfolio and the risk-free asset
- In equilibrium, demand equals supply:

The tangency portfolio of all securities equals the market portfolio

Systematic and Idiosyncratic Risks

Systematic Risk

- Systematic risk is risk to which <u>all</u> firms are exposed
- Apple examples: sales fall when unemployment rises, wages fall; conversely, sales rise when unemployment falls, wages rise
- Home Depot is exposed to the same risks

Idiosyncratic Risk

- Idiosyncratic risk is risk specific to the firm
- Apple examples: problems with Foxconn (an Apple supplier), Steve Jobs dies, the Justice Department figures out how to hack iPhones, etc.
- Home Depot is not exposed (directly) to these risks

The CAPM in a Nutshell

- In a nutshell, the CAPM says that

$$\overbrace{R_{\text{Stock}} - R_F}^{\text{Stock Return}} = \overbrace{\beta_{\text{Stock}} \times (R_{\text{Market}} - R_F)}^{\text{Systematic Risk}} + \overbrace{\epsilon_{\text{Stock}}}^{\text{Idiosyncratic Risk}}$$

  where ϵ_{Stock} is a random shock to the stock's return
- **Important:** $\text{E}[\epsilon_{\text{Stock}}] = 0$ and $\text{Cov}[R_{\text{Market}}, \epsilon_{\text{Stock}}] = 0$
- Taking expectations,

$$\overbrace{E[R_{\text{Stock}}] - R_F}^{\text{Stock Risk Premium}} = \beta_{\text{Stock}} \times \overbrace{(E[R_{\text{Market}}] - R_F)}^{\text{Equity Risk Premium}}$$

- Investors are <u>not</u> compensated for idiosyncratic risk: the stock's risk premium only depends on its beta (β_{Stock}) and the equity risk premium (a.k.a. market risk premium)

CAPM Assumptions

1. **Investor behavior**
   1.1 Investors are rational, mean-variance optimizers
   1.2 Their common planning horizon is a single period
   1.3 Investors all use identical input lists, and assumption often termed homogeneous expectations. Homogeneous expectations are consistent with the assumption that all relevant information is publicly available
2. **Market structure**
   2.1 All assets are publicly held and trade on public exchanges
   2.2 Investors can borrow or lend at a common risk-free rate, and they can take short positions on traded securities.
   2.3 No taxes.
   2.4 No transaction costs.

Beta

$$\overbrace{R_{\text{Stock}} - R_F}^{\text{Stock Return}} = \overbrace{\beta_{\text{Stock}} \times (R_{\text{Market}} - R_F)}^{\text{Systematic Risk}} + \overbrace{\epsilon_{\text{Stock}}}^{\text{Idiosyncratic Risk}}$$

- β_{Stock} (Beta) is the slope of the regression line obtained by regressing the stock's returns on the market's returns

$$\beta_{\text{Stock}} = \frac{\text{Cov}[R_{\text{Stock}}, R_{\text{Market}}]}{\text{Var}[R_{\text{Market}}]} \tag{1}$$

- The beta of a security is the expected % change in its return given a 1% change in the return of the market portfolio
- It is the sensitivity of the stock's returns to systematic risk

CAPM Beta: Examples

Walmart Inc. (WMT) Add to watchlist

NYSE - NYSE Delayed Price. Currency in USD

96.90 +1.07 (+1.12%) At close: 4:00PM EDT

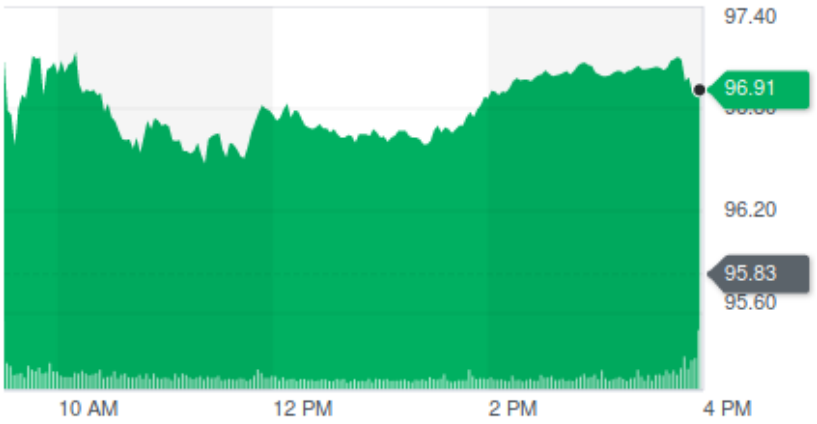
96.95 +0.05 (0.05%) After hours: 6:43PM EDT

Buy Sell

Summary | Chart | Conversations | Statistics | Profile | Financials | Options | Holders | Historical Data | Analysis | Sustainability

Previous Close	95.83	Market Cap	283.794B
Open	96.11	Beta	0.25
Bid	0.00 x 1100	PE Ratio (TTM)	55.43
Ask	0.00 x 3000	EPS (TTM)	1.75
Day's Range	96.09 - 97.19	Earnings Date	Nov 15, 2018
52 Week Range	77.50 - 109.98	Forward Dividend & Yield	2.08 (2.16%)
Volume	8,332,605	Ex-Dividend Date	2018-12-06
Avg. Volume	7,814,630	1y Target Est	104.00

1D 5D 1M 6M YTD 1Y 5Y Max Full screen



Walmart has a beta of **0.25**

CAPM Beta: Examples

Tiffany & Co. (TIF) Add to watchlist

NYSE - NYSE Delayed Price. Currency in USD

124.32 +1.45 (+1.18%) At close: 4:02PM EDT

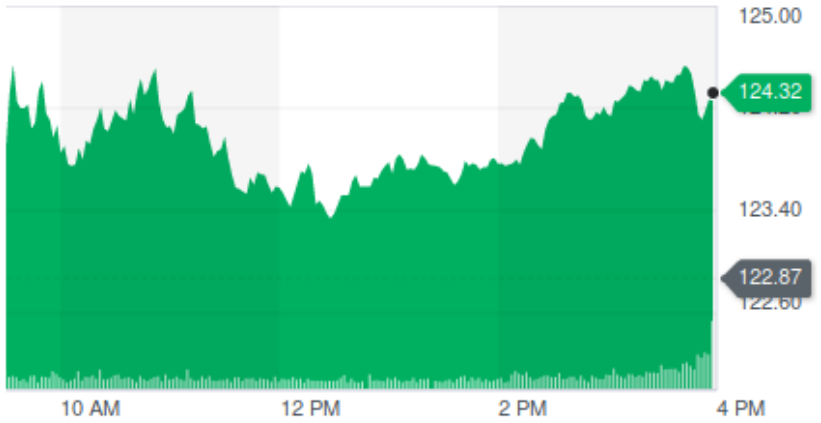
124.09 -0.23 (-0.19%) After hours: 4:26PM EDT

Buy Sell

Summary | Chart | Conversations | Statistics | Profile | Financials | Options | Holders | Historical Data | Analysis | Sustainability

Previous Close	122.87	Market Cap	15.218B
Open	123.86	Beta	1.47
Bid	123.40 x 800	PE Ratio (TTM)	34.46
Ask	124.76 x 1400	EPS (TTM)	3.61
Day's Range	123.29 - 124.80	Earnings Date	Nov 27, 2018 - Dec 3, 2018
52 Week Range	86.15 - 141.64	Forward Dividend & Yield	2.20 (1.78%)
Volume	1,498,549	Ex-Dividend Date	2018-09-19
Avg. Volume	1,598,087	1y Target Est	140.50

1D 5D 1M 6M YTD 1Y 5Y Max Full screen



Tiffany's has a beta of **1.47**

Volaility Decomposition

- The stock's volatility is $\mathrm{SD}[R_{\mathrm{Stock}}]$
- The volatility of a porfolio with the same systematic risk as that of the stock is $\beta_{\mathrm{Stock}}\mathrm{SD}[R_{\mathrm{Market}}]$
- We say that a fraction

$$\frac{\beta_{\mathrm{Stock}}\mathrm{SD}[R_{\mathrm{Market}}]}{\mathrm{SD}[R_{\mathrm{Stock}}]} \tag{2}$$

  of the stock's volatility is systematic; we say that the remaining fraction is idiosyncratic
- Note that Equation 2 is just $\sqrt{R^2}$ from regression analysis

Example

Consider the following data for Cisco (CSCO) and the market

$$\begin{array}{ll} E[R_{CSCO}] = 0.9961\% & SD[R_{CSCO}] = 7.09\% \\ E[R_{MKT}] = 1.0773\% & SD[R_{MKT}] = 3.7568\% \\ Corr[R_{CSCO}, R_{MKT}] = 0.6339 & R_F = 0.0418\% \end{array}$$

1. What are Cisco's and the market's Sharpe ratios? .1346, .2756
2. What is Cisco's beta? 1.1963
3. What is the market's risk premium? 1.0355%
4. What is Cisco's risk premium? 0.9543%
5. What fraction of Cisco's volatility is systematic? .6339
6. What fraction is idiosyncratic? .3661

Example

Consider the following data for Caterpillar (CAT) and the market

$$\begin{array}{ll} \mathrm{E}[R_{\mathrm{CAT}}] = 1.3424\% & \mathrm{SD}[R_{\mathrm{CAT}}] = 8.1365\% \\ \mathrm{E}[R_{\mathrm{MKT}}] = 1.0773\% & \mathrm{SD}[R_{\mathrm{MKT}}] = 3.7568\% \\ \mathrm{Corr}[R_{\mathrm{CAT}}, R_{\mathrm{MKT}}] = 0.7279 & R_F = 0.0418\% \end{array}$$

1. What are Caterpillar's and the market's Sharpe ratios?
2. What is Caterpillar's beta
3. What is the market's risk premium?
4. What is Caterpillar's risk premium?
5. What fraction of Caterpillar's volatility is systematic?
6. What fraction is idiosyncratic?

Alpha

- The difference between the stock's actual risk premium and the risk premium of a portfolio with the same systematic risk is the alpha:

$$\alpha_{\text{Stock}} = \underbrace{(\mathrm{E}[R_{\text{Stock}}] - R_F)}_{\text{True Value}} - \underbrace{\beta_{\text{Stock}} \times (\mathrm{E}[R_{\text{Market}}] - R_F)}_{\text{CAPM}} \quad (3)$$

- If $\alpha \neq 0$, then one can make an arbitrage profit
  - If $\alpha > 0$, then go long
  - If $\alpha < 0$, then go short
- While one might observe a non-zero alpha in the short-term, it will likely become zero in the long-term

Example

1. What is Cisco's alpha? -0.2845%
2. Should we go long or short? short
3. What about Caterpillar?